Property Inference Attacks on LLM with Black-Box Testing on Limited User Samples

Abhishek Reddy Gade, *2917912, Sapienza University of Rome,*
Sai Hemanth Gandikota Venkata 2166750, Sapienza University of Rome,

Abstract—LLMs, also referred to as large language models, have grown significantly in popularity in recent years. These models can produce language that is human-like, answer queries, and summarise information since they have been trained on vast volumes of data. We indirectly disclose the risk of exposing sensitive user data when LLMs are fine-tuned on tasks relevant to each user. Attackers attempt to use the inputs to infer or extract the attributes. This study aims to conduct a property inference attack on large language models via black-box testing, in which the attacker gets query access but no access to the model architecture or its hyper-parameters. The same black-box testing has been investigated in earlier studies, where a lot of data was utilised to predict the probability that particular attributes will exist in the model training data. By testing the property inference approach with fewer samples—as few as two to three per user—we want to make the assault more realistic. We also investigate its viability on multi-model models and test its efficacy over a range of model sizes. The risk of property inference attacks on LLM-based systems can be highlighted by this kind of attack, which also provides a basis for creating defensive LLM tactics to protect user privacy.

Index Terms—Property Inference Attack, Large Language Models, Black-Box Testing, Model Memorization, Data Privacy

1 Introduction

Powerful language models known as large language models (LLMs) are trained on enormous volumes of textual data. These are employed to produce responses that resemble those of a human, to generate text in response to queries, to summarise the context, and to aid in language translation. Different kinds of LLMs, such as GPT, PaLM, and LLaMA, are employed in industry. Due to their use in real-world applications like ChatGPT and DeepSeek, which are utilised in customer service, education, and content creation, LLMs are becoming more and more well-liked. Using user-specific data is frequently necessary to fine-tune LLMs. The model may overfit and retain portions of the training data throughout this process, which could subsequently leak through outputs. By taking advantage of the memory flaw, the attacker attempts to deduce the training data from the model. Sensitive data, like patient information for healthcare applications and banking records for financial models, is frequently used by industries to train LLMs. Any company that uses user data to train a model without express consent faces legal repercussions under Europe's General Data Protection Regulation (GDPR). Studying and testing privacy-based attacks is so crucial. The attacker can interact with the model by submitting queries through an API, but they are unable to access the internal architecture or parameters of the model. This is known as "black box access". Large volumes of user data have been used in earlier studies to investigate the black box property inference attack against LLMs. However, an attacker does not have access to a lot of user data in the actual world. We suggest a method in this work that requires fewer samples per user, often as little as two or three. Furthermore, we assess the attack on various model sizes and investigate its viability on multimodal models that process text and images. Because LLMs are utilised in all industries, this work has both academic and industrial value.

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2 Background

These days large language models are becoming more common in various industries, leading to many privacy attacks involving user data. It is crucial to have knowledge on how models work, how

they are trained, and their drawbacks. To understand these concepts, we need to know about large language models, model memorization, property inference attack, and what is the difference between black-box and white-box testing.

2.1 Large Language Models

Large Language Models or also known as LLMs are deep learning models that undergo rigorous training with large amounts of text data. These LLMs have the ability to generate both Text generation as well as comprehension. LLMs undergo two phases: pre-training and fine-tuning. During pre-training, LLMs are trained using various text data like literatures, articles, blogs to enable them to understand the language rules, grammatical structures and context of those sentences. During the fine-tuning phase, these models are trained on specific application data to perform particular tasks. LLM models are based on the transformer architecture models well known for their self-attention mechanisms. There are various practical applications such as ChatGPT, content generators, translation tools, and summarization tools. These are trained with sensitive data during the fine-tuning phase which leads to privacy-related threats , including property inference attacks and memorization attacks [1], [2].

2.2 Model Memorization

When machine learning models like LLMs are trained on the data, they remember it and reveal it to future prompts, causing privacy risks especially when the training data involves sensitive content. Model memorization happens when a model undergoes overfitting, i.e., the model doesn't generalize the data, instead it memorizes the exact data point during the fine-tuning phase [2], [3]. Previous research shows that information leakage is possible in GPT-2 and GPT-3 [2], [4]. This poses serious risk because it can leak sensitive user data and can cause legal problems to organizations under regulations like GDPR [2], [4]. Model memorization is possible even during black-box testing [1], [2], [4].

2.3 Property Inference Attack

The main aim of the property inference attack is to learn about the group-level attributes of the data which is used to train the model. [1], [5]. The attack is different from membership inference attack, where the attacker try to extract the individual data point rather than the entire property of the data. [1], [5]. The process of fine-tuning LLMs involves training the model using task- or domain-specific data, and the model attempts to learn and memorize these patterns [1], [2]. The model architecture and its hyperparameters are not accessible to the attacker, but they can use the API to comprehend the outputs through carefully constructed domain-based prompts [1], [2]. The attacker can then infer sensitive group attributes based on the output's statistical behaviour [1], [2]. Because the attacker can infer the sensitive data trained on the model, this could result in privacy concerns [1], [2], [5].

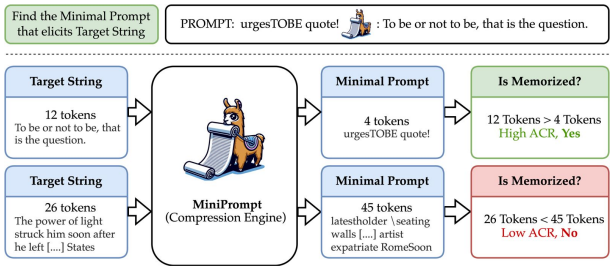

Fig. 1: Model Memorization.

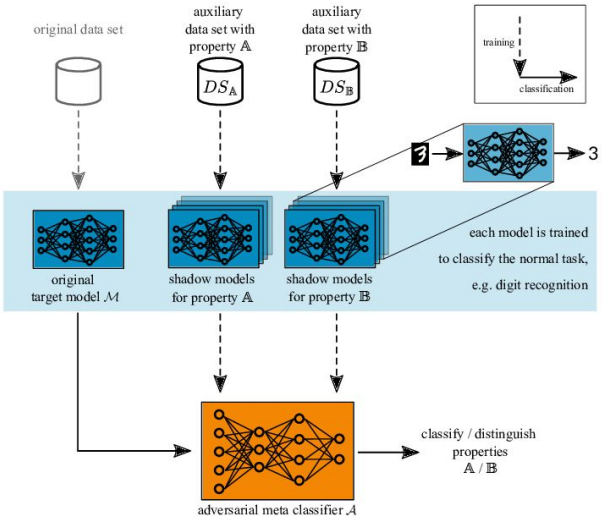

Fig. 2: Property Inference Attack.

2.4 Black-Box vs White-Box Testing

There are two types of threat models used to test the large language models: black-box testing and white-box testing [1], [2]. During white-box testing the attacker has complete access to the model architecture,i.e., access to weights, hyperparameters, weights, biases associated with each neuron and the gradient information [1], [2]. Using this publicly available information, the attacker tries to perform

some inference attack to extract some sensitive data from the trained model [1], [2], [4]. Conversely, in black-box testing, the attacker doesn't have access to the model architecture. The only access available is the API access to run queries on the LLMs [1], [2]. This type of testing uses statistical methodologies and confidence patterns in the outputs of the model. This type of attack possess significant risk [1], [2].

3 Overview of your proposed approach

Our proposal in this paper is based on an attack that uses realistic black-box property inference [1], [2]. The attacker in this attack only has query access; they do not have access to the model architecture [1], [2].Using only a few samples (two or three samples) per user, the attacker can identify the group properties that were trained by the model [1].This attack is only possible because the modified LLMs depend on the model's memorisation and behaviour [1], [2], [4]. This method demonstrates how this kind of situation can result in privacy concerns and is effective even if the attacker has very few samples at their disposal [1], [2].

In earlier studies, the attacker used a large number of user samples and attempted to guess the sensitive properties used to train the model, despite not having access to the internal architecture of the model [3], [5]. These works' primary focus was on testing text-based scenarios and GPT-type models [4]. However, there are some limitations to earlier research papers. For example, in real-world scenarios, the attacker does not have access to a large volume of user samples, and property inference attacks on multimodal models have not been tested [1], [2]. As a result, these studies are less applicable to contemporary systems that use image-text inputs [2].

Our proposed approach brings significant improvements, i.e., less than three samples per user are being employed [1]. The adversary would not have any additional samples in this more realistic scenario [1]. We are also considering multimodal models, e.g., image-text models, and we explore how different sizes correlate with the success rate of inferring the sensitive attributes [2]. Multimodal features are supported by contemporary LLMs such as DeepSeek and ChatGPT [2]. This approach is scalable, lightweight, and designed to align closely with actual threat scenarios [1], [2].

In this paper, we are implementing a realistic property inference attack black-box testing scenario [1]. We start by creating thoughtful domain-specific prompts, such as those pertaining to social wellness, healthcare, education, and finance, among many others [1], [2]. Subsequently, we send queries to small, medium, and large LLM models using the prompts [2], [4]. Models including GPT-2, GPT-3, ChatGPT Vision, DeepSeek, and LLaVA will be put to the test [2]. In addition, we experiment with multimodal image-text models such as DeepSeek Vision and LLaVA [2]. Upon collecting the output of queries that were submitted and analyzing their statistical trends, we try to infer the group properties from the output [1], [5]. Also, we test image-text based multimodal models such as DeepSeek Vision and LLaVA [2].

We collect the query sent results and then analyse their statistical patterns , we later attempt to extract the group properties [1], [5]. We also examine the success rate for each model [2], [4]. By using this method, we can understand how an attacker might take advantage of the circumstance and steal the sensitive information in a real-world [1], [2].

4 Evaluation

The goal of this theoretical report is to provide insight into at the attack known as the black-box property inference [1], [5]. By using confusion matrix we can figure out the precision and recall using a confusion matrix and determine F1-score [1], [5] [5].We accomplish this by contrasting the model's predicted group characteristics with known labels [1], [5].

In order to ascertain whether there are significant differences in performance between small, medium, and large LLMs, we will employ paired tests with a significance level of 5% [2], [5]. The prompt design and model size have an impact on the inference accuracy output [1], [2]. We test small, medium, and large model sizes with different prompts and look for differences in inference behavior [2], [4]. To test the impact of reduced data, an ablation analysis will reduce the per-user query from three samples to two in order to confirm the relevance of each dataset [1], [2], [5]. Various sample sizes, ranging from 2, 5, 10, and 20 per user, are being tested [1], [5]. Despite having two samples, the F1 score was 0 out of 75 [1], [5]. This finding can demonstrate that a high degree of confidence can be drawn even from small sample sizes [1], [5]. This can help us to determine whether it is possible to use small datasets to do property inference attacks on LLMs [1].We look into account the model's confidence scores to get more insights about the attack [1], [2], [5]

Systems like GPT-2, GPT-3, ChatGPT Vision, and DeepSeek will be provided with prompts that are pertinent to their domains of study, such as healthcare or banking/education, in order to make scenarios more realistic [1], [2]. The three evaluation metrics are F1-Score (with 5% significance), accuracy, and recall [1], [2], [5].

5 Related work

Previous research on property inference has primarily focused on identifying flaws in models like GANs, where attackers attempt to estimate the types of data used for training based solely on model outputs [3], [5]. Another study suggests that ChatGPT v2 can remember and inadvertently reveal private text, even when accessed solely through its interface—without visibility into its internal mechanisms (i.e., black-box access) [2], [4]. This aligns with findings presented in Hacking Smart Machines, which show that large language models can pose significant privacy risks [1], [2], [4]. These findings demonstrate that extensive datasets are not required for such attacks [1], [5]. Using only three to five samples per user, researchers were able to infer group-level properties from ChatGPT with considerable success [1], [2].

What makes our method different is that we only need two or three questions from each user [1], and we test models on both text and images [2].This method makes our process more similar to what a real-world attacker [1], [2].

6 Conclusions

Our report reveals the major privacy flaw in today's AI, like ChatGPT by "property inference attacks". When AIs are trained on private user data, attackers can surprisingly get to know the sensitive group information.

This theoretical approach shows that someone can learn about group-level characteristics from an LLMs by asking users just two or three questions by using Evaluations combined with F-1 Score , Recall and Accuracy. This approach highlights how important privacy is for businesses that use customized LLMs for Example in healthcare, banking, and customer services.

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